NEW HORIZON COLLEGE OF ENGINEERING
New Horizon Knowledge Park, Ring Road, Marathalli
Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
Accredited by NAAC with 'A' Grade, Accredited by NBA

Department of Artificial Intelligence & Machine Learning

Mini Project -(24AIM57)

Academic Year-2026-27

PROJECT SYNOPSIS

Students Details:

Sl. No.	USN	Name	Sem	Mobile Number	Email Id	Signature
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Project Title :

Al-Based Crop Recommendation for Farmers

Introduction:
Farmers often find it hard to decide which crop to grow. The right crop depends on soil nutrients, rain, temperature, and more. This project uses Artificial Intelligence (AI) to look at these things and suggest the best crops for a farmer's land. The farmer gets the answer in a simple app

Need / scope of the project:
Many farmers pick crops by habit or guesswork. A wrong choice can cause low yield and money loss.
Soil testing reports are hard to understand for many people.
Weather is becoming less predictable, so old methods may not work well.
The project will cover Karnataka crops first and can grow to other states later.
The app will give the top 3 crops, the reason for each, and expected profit.

Methodology to be used:

1. **Collect data:** Soil data (N, P, K, pH), weather data (temperature, humidity, rainfall), and crop data from public sources like Kaggle, ICAR, and data.gov.in.
2. **Clean the data:** Remove errors and make all values follow the same scale.

3. **Train models:** Use machine learning models like Random Forest and XGBoost , and pick the one with the best accuracy.
4. **Explain the result:** Use SHAP charts to show why a crop was chosen.
5. **Add live weather and prices:** Use a free weather API and market (mandi) prices to rank crops by profit.
6. **Build an app:** Make a simple Streamlit app where the farmer enters details and gets a suggestion. Add Kannada, Telugu and Hindi language options.

Expected outcomes:

A trained AI model that suggests the best crop with high accuracy.
A working app that shows the top 3 crops with reasons.
Charts that explain the model's decisions in an easy way.
Use Deep learning models as well for better decisions and predictions for farmers.
A project report with test results and a comparison of different models.
Being helpful to Farmers in South India primarily and then extending into the Northern states

Societal or any other benefits:

Helps farmers choose the right crop, so they can earn more and waste less.
Saves water, fertilizer, and money by growing crops that suit the land.
Reduces the risk of crop failure.
Local language support helps farmers who are not comfortable with English.
Supports smart and sustainable farming, which helps food security.
There are countless farmers who die of starvation / loans / many other troubles ,
Farmers are the backbone of our nation so this aims to better their lives and help them in their profession.

Guide Details:

Sl. No.	Faculty Name	Designation	Department	Signature
1	Dr. Sanjeev P Kaulgud	Associate-Professor	AIML	